Quantifying Epistemic Uncertainty in Diffusion Models

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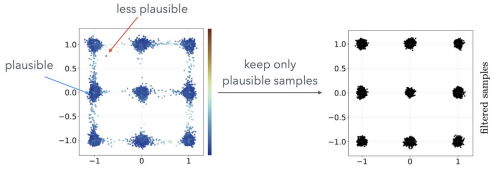
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UNCERTAINTY IN DIFFUSION MODELS

Diffusion models are used in a broad range of applications, from image generation to scientific applications, where they generate time series, molecules, materials, and proteins.

But they do not know when they do not know: While probabilistic by design, random variation is not the same as uncertainty. Their randomness is mostly sampling noise and not a reliable measure of whether the diffusion model has learned the right thing.

A simple example: With nine Gaussian modes arranged on a square grid, samples between the modes are actually implausible. Reliable uncertainty estimates help us to filter out these samples.



Two types of uncertainty: Aleatoric noise comes from the randomness of generation, while epistemic uncertainty comes from what the model does not know. Existing UQ methods for diffusion models often mix the two, making it hard to tell whether a sample is actually plausible or just produced with confidence.

Total Predictive Uncertainty

$$\underbrace{\mathrm{Var}[x]}_{\text{total}} = \underbrace{\mathbb{E}_\theta[\mathrm{Var}(x|\theta)]}_{\text{aleatoric}} + \underbrace{\mathrm{Var}_\theta[\mathbb{E}(x|\theta)]}_{\text{epistemic}}$$

Captures both data randomness and model uncertainty.

Our contribution: We derive a scalable Fisher–Laplace projection that pushes parameter uncertainty through the denoising trajectory, isolating epistemic variance from diffusion noise. This enables us to quantify uncertainty and use it to filter out implausible samples.

METHODOLOGY

Proposition: Assuming a local Gaussian approximation of the parameter posterior, we can decompose the covariance at each reverse diffusion step into two parts: uncertainty from the model parameters and noise from the sampling process.

One-step Fisher–Laplace Projection

$$\boldsymbol{\Sigma}^{\mathrm{ep}}_{t-1|t}(\eta) = b_t^2\, \boldsymbol{J}_t\, \boldsymbol{\Sigma}_\theta\, \boldsymbol{J}_t^\top .$$

Multi-step Epistemic Recursion

$$\boldsymbol{\Sigma}^{\mathrm{ep}}_{t-1}(\eta) = a_t^2\, \boldsymbol{\Sigma}^{\mathrm{ep}}_t(\eta) + b_t^2\, \boldsymbol{J}_t\, \boldsymbol{\Sigma}_\theta\, \boldsymbol{J}_t^\top .$$

Algorithm 1: FLARE

Inputs : diffusion model ε_θ; subnetwork size m; Hessian–vector products for $(\boldsymbol{H} + \lambda \mathbf{I})$; DDPM schedule $\{\beta_t\}_{t=1}^T$; $\boldsymbol{x}_T \sim \mathcal{N}(0, \mathbf{I})$.

Outputs: generated sample $\hat{\boldsymbol{x}}_0$; epistemic covariance $\boldsymbol{\Sigma}^{\mathrm{ep}}(\hat{\boldsymbol{x}}_0)$.

Precompute: $\alpha_t \leftarrow 1 - \beta_t$;
$\bar{\alpha}_t \leftarrow \prod_{s \le t} \alpha_s$;
$a_t \leftarrow \alpha_t^{-1/2}$;
$b_t \leftarrow \beta_t / (\sqrt{\alpha_t}\sqrt{1-\bar{\alpha}_t})$.

Sample subnetwork (once): draw $I \subset [p]$;
define solve operator $\boldsymbol{\Sigma}_{\mathrm{sub}} v \coloneqq (\boldsymbol{H}_{I,I} + \lambda \mathbf{I})^{-1} v$

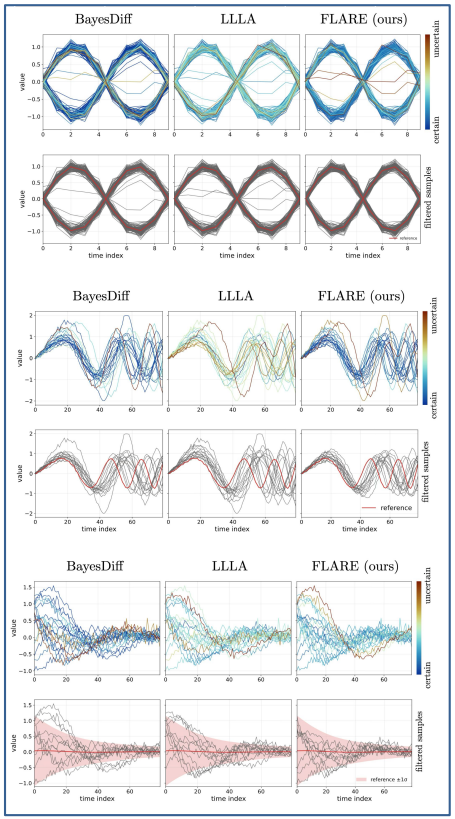
Initialize: $\hat{\boldsymbol{x}}_T \leftarrow \boldsymbol{x}_T$;;
$\boldsymbol{\Sigma}^{\mathrm{ep}}(\hat{\boldsymbol{x}}_T) \leftarrow \mathbf{0}$;

for $t = T$ **to** 1 **do**

- $\hat{\varepsilon}_t \leftarrow \varepsilon_\theta(\hat{\boldsymbol{x}}_t, t)$;
- $\hat{\boldsymbol{x}}_{t-1} \leftarrow a_t\left(\hat{\boldsymbol{x}}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\hat{\varepsilon}_t\right)$;
- $\boldsymbol{J}_{t,I} \leftarrow \nabla_{\theta_I} \varepsilon_\theta(\hat{\boldsymbol{x}}_t, t)$;
- $\Delta_t \leftarrow \boldsymbol{J}_{t,I}\, \boldsymbol{\Sigma}_{\mathrm{sub}}\, \boldsymbol{J}_{t,I}^\top$;
- $\boldsymbol{\Sigma}^{\mathrm{ep}}(\hat{\boldsymbol{x}}_{t-1}) \leftarrow a_t^2\, \boldsymbol{\Sigma}^{\mathrm{ep}}(\hat{\boldsymbol{x}}_t) + b_t^2\, \Delta_t$;

return $\hat{\boldsymbol{x}}_0,\ \boldsymbol{\Sigma}^{\mathrm{ep}}(\hat{\boldsymbol{x}}_0)$

QUALITATIVE RESULTS



QUANTITATIVE RESULTS

Table 1: Comparison of uncertainty-based filtering.Higher gap-closure and lower ROC-AUC indicate improved alignment between filtered generations and the training data.

Dataset	Method	Gap-Closure (%)	ROC-AUC (↓)	p (bootstrap)
Sines	BayesDiff	+13.3654	0.6153	0.0001
	LLLA	+47.1873	0.5814	0.0201
	FLARE (ours)	**+93.0814**	**0.5003**	**0.0012**
Chirp	BayesDiff	+41.7337	0.6616	0.0030
	LLLA	+59.0583	0.5891	0.0116
	FLARE (ours)	**+74.3137**	**0.5345**	**0.0002**
Damped Sines	BayesDiff	+10.3984	0.6861	0.0001
	LLLA	−18.7540	0.7754	0.0066
	FLARE (ours)	**+85.0**	**0.5085**	**0.0002**

THEORETICAL GUARANTEE FOR FLARE

Theorem 1. *Let $\boldsymbol{J}_{\mathrm{pop}} \in \mathbb{R}^{nd \times p}$ and $\boldsymbol{J}_t \in \mathbb{R}^{d \times p}$ satisfy the assumptions above. Then, with probability at least $1-\delta$, the randomized subnetwork estimator with subnetwork size m satisfies*

$$\left|\mathrm{tr}(\boldsymbol{V} - \tilde{\boldsymbol{V}})\right| \le \tilde{\mathcal{O}}\left(\sqrt{\frac{p\,\gamma\,\mu}{m\,\delta}}\,\kappa\right)\mathrm{tr}(\boldsymbol{V}).$$

CONCLUSION & FUTURE WORK

We demonstrated the advantage of our approach for uncertainty quantification on synthetic time-series benchmarks designed to expose specific epistemic failure modes.

Although randomized sampling reduces computational cost, estimating curvature in very large diffusion models remains challenging. Future work could improve scalability through structured subsampling or low-rank curvature approximations.